Explainable Artificial Intelligence (XAI)

An introduction

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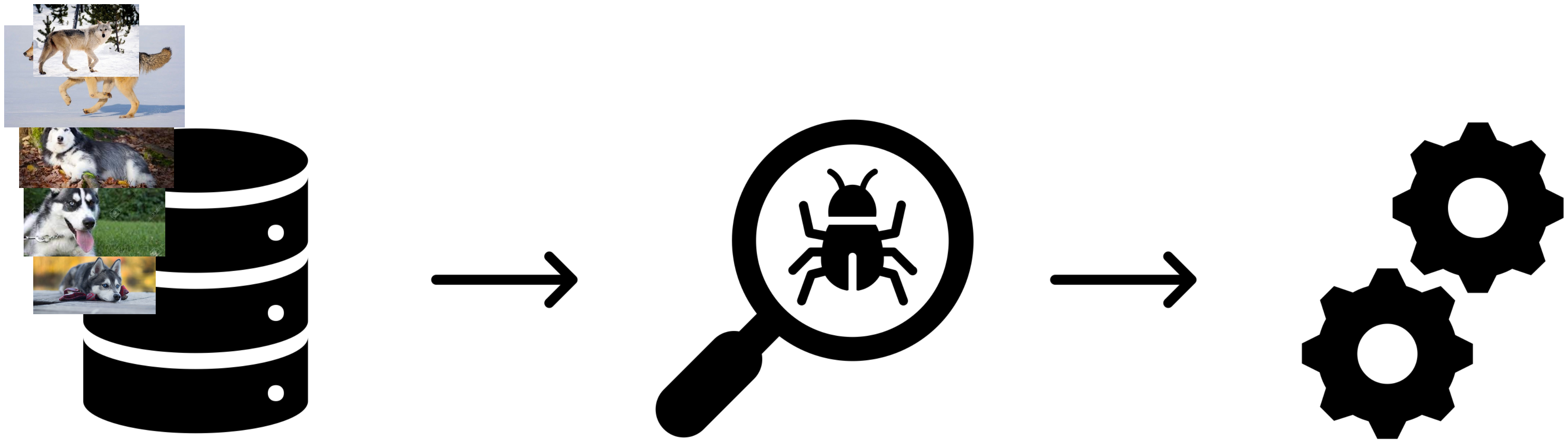
2025-02-06, Karlskrona

Classify huskies and wolves

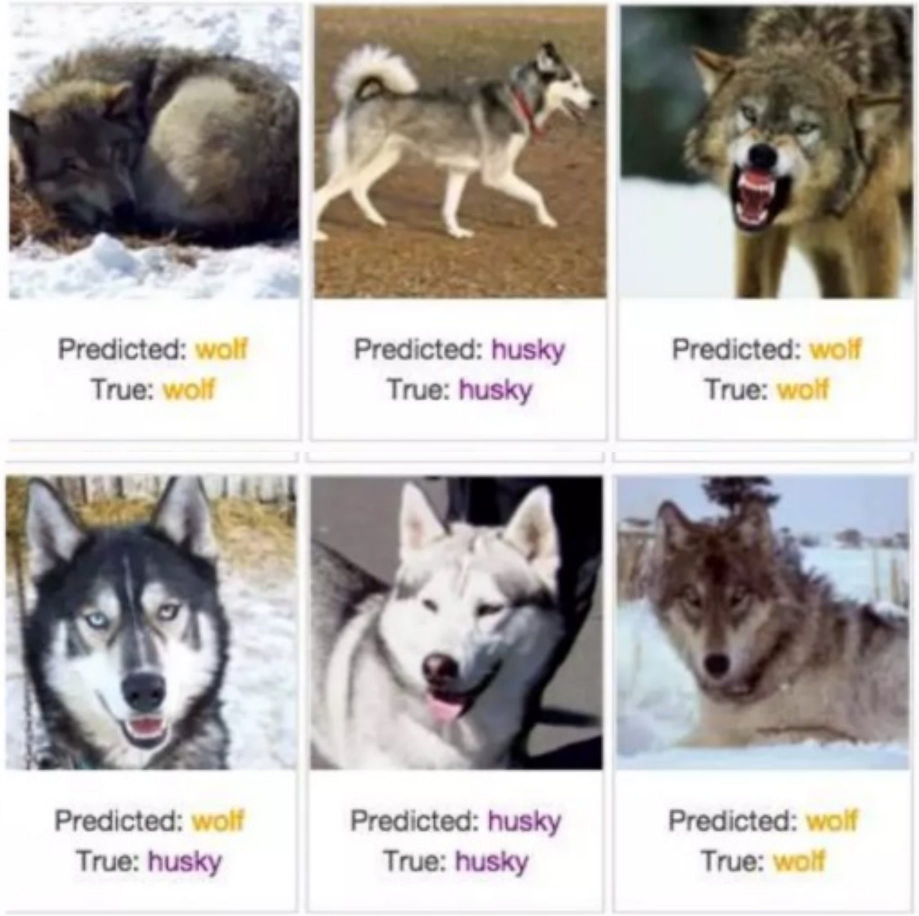
Differences?

Train a model



Data Training Model

Predictions

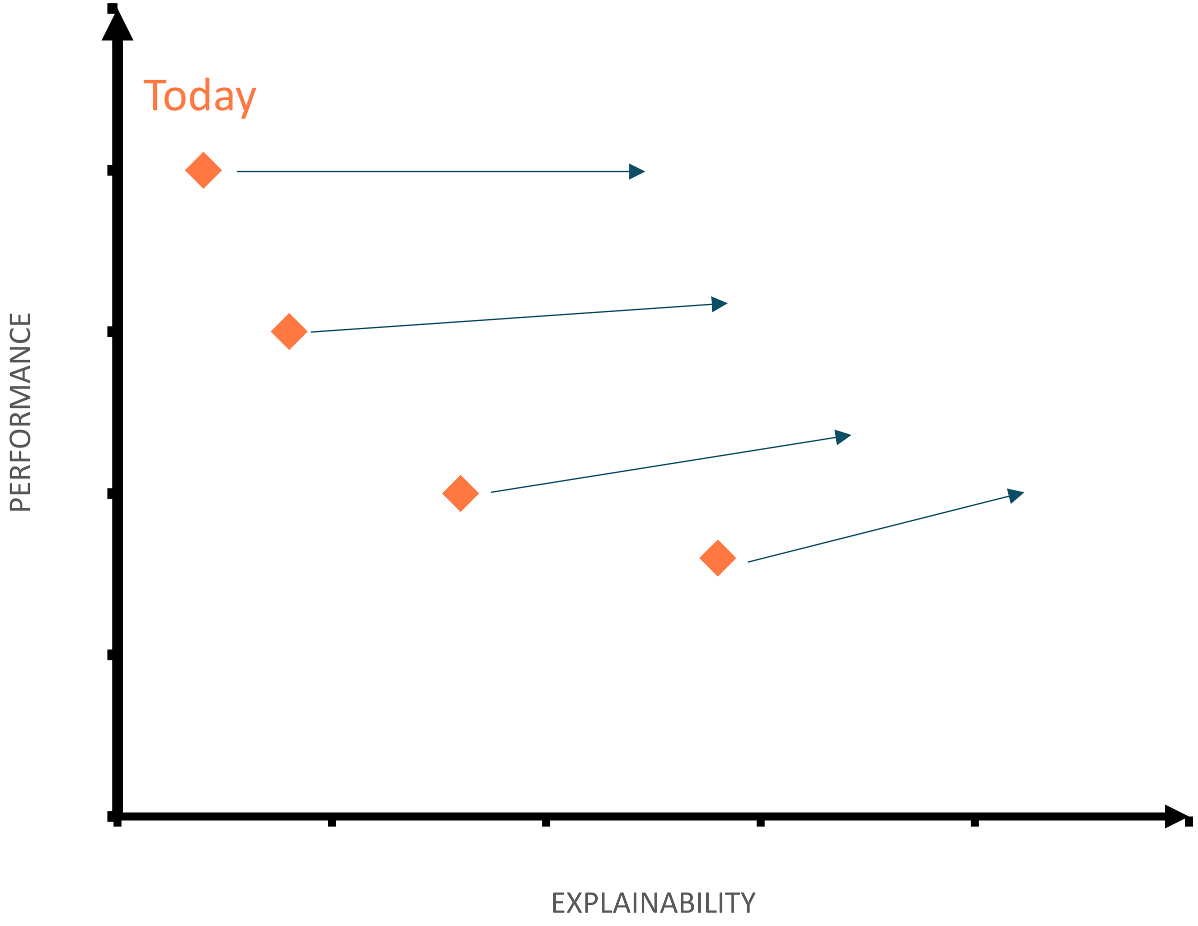


What is XAI?

Core concepts

Explainable AI

- Methods and models that make the model and its predicitons understandable to humans
- We want to trust and rely on models
- Explainability and performance contradictive
  - Hopefully we get better!



Core concepts
Intrinsic vs Post hoc

Intrisically Interpretable Models

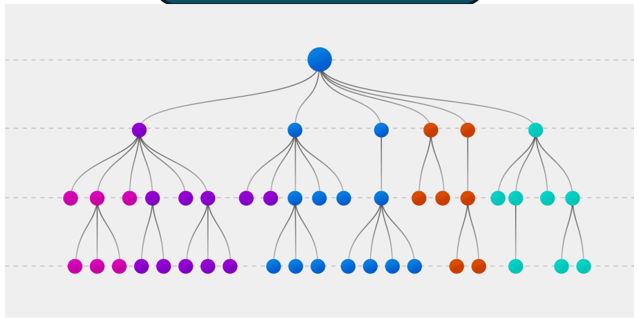
- The model is the explanation

Post hoc methods

- Methods that analyze after training
- Usually analyzes by comparing input and output pairs

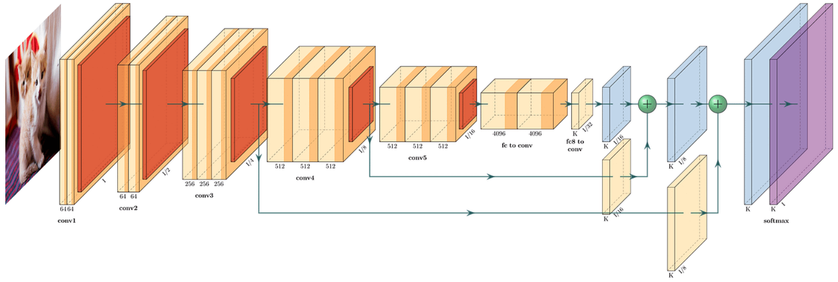
XAI

Intrinsically explainable



Linear models
Decision trees
Rule-based models

Post-hoc explainable



Deep Neural Networks
Ensemble models

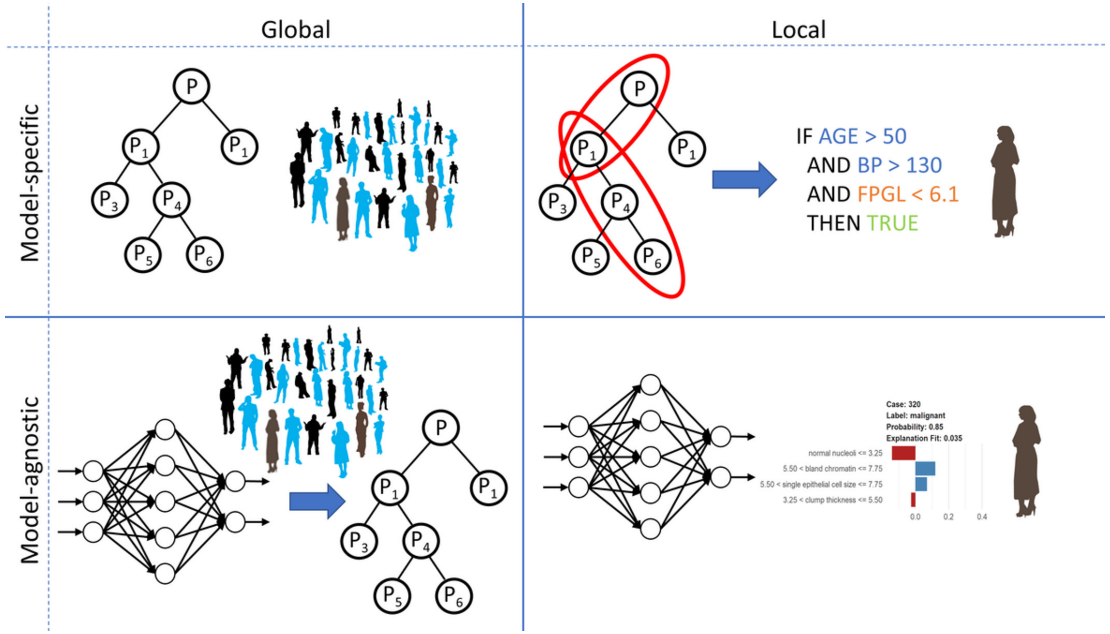
Core concepts
Interpretability Scope

Global

- Entire model
- Which features are important
- What interactions are there between the features
- Difficult to achieve in practice

Local

- Individual instances/predictions
- Reduces complexity



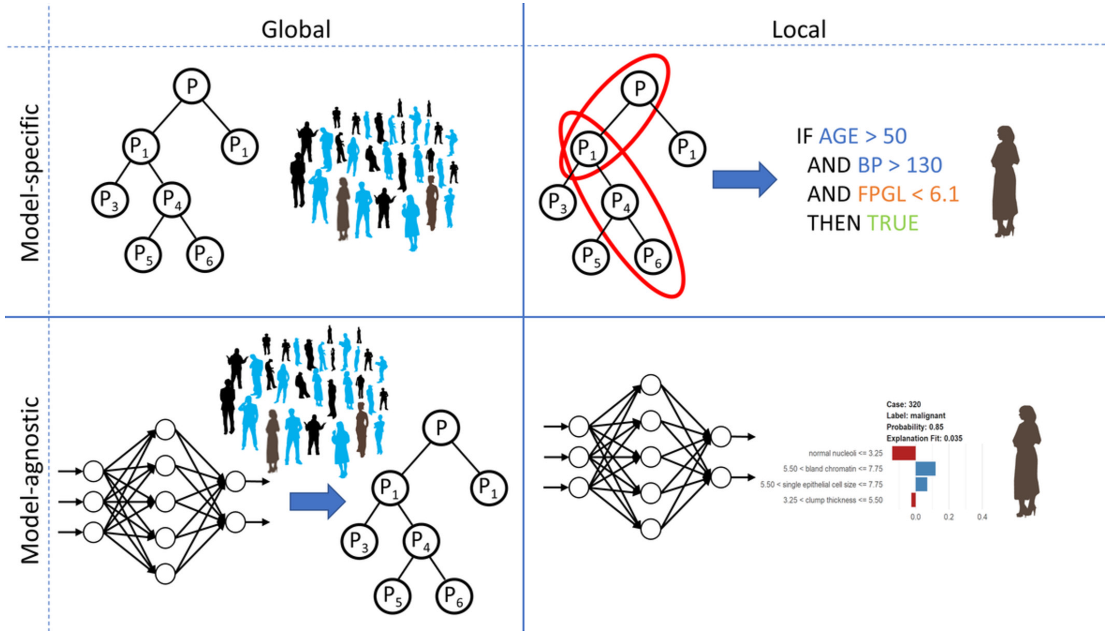
Core concepts
Model-Agnostic vs Model-Specific

Model-Specific

- Limited to specific models

Model-Agnostic

- Tools that can be used on any model, including the intrisically interpretable
- Applied after model has been trained
- Usually analyzes by comparing input and output pairs



Core concepts
Evaluation of Interpretability

No concensus on how to do it… But, we dont want just another score to look at.

Application level (real task)

- Have the end user test it
    - Radiologist using a fracture detection software

Human Level (simple task)

- Have some laymen test it
- Give multiple explanations, let user select the best one
    - Me using a fraction detection software

Function level (proxy task)

- Some criterion that does not require human input
    - Depth of resulting decision tree

Interpretable Models

Interpretable Models
Linear Regression

Defined as

$$y = \beta_0 + \boxed{\beta_1} x_1 + \ldots + \boxed{\beta_p} x_p + \epsilon$$

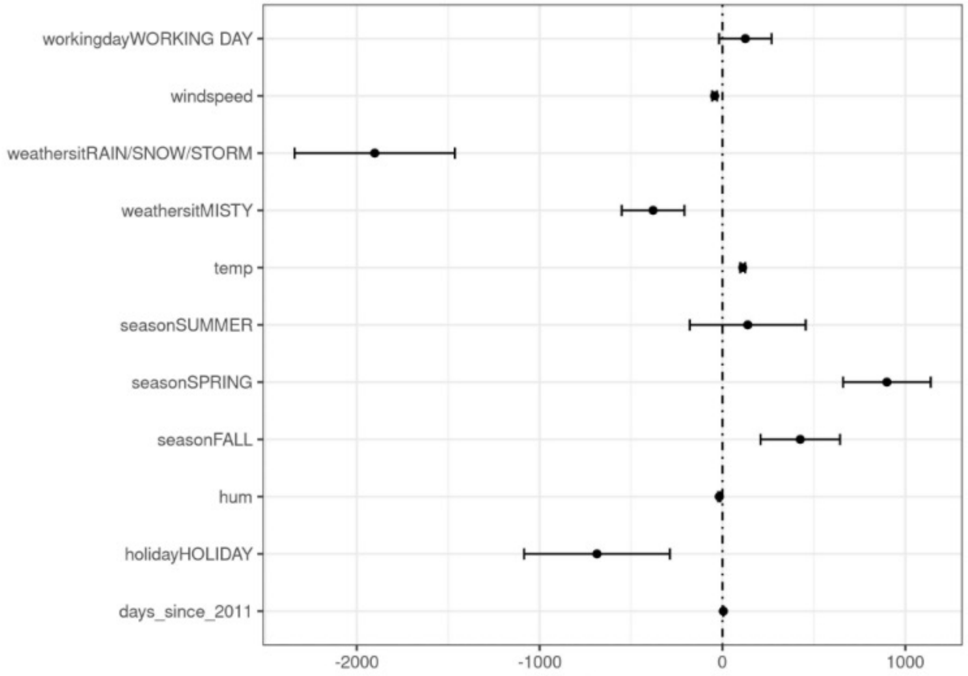
Feature importance can be measured by the absolute value of its t-statistic

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

Can see it as

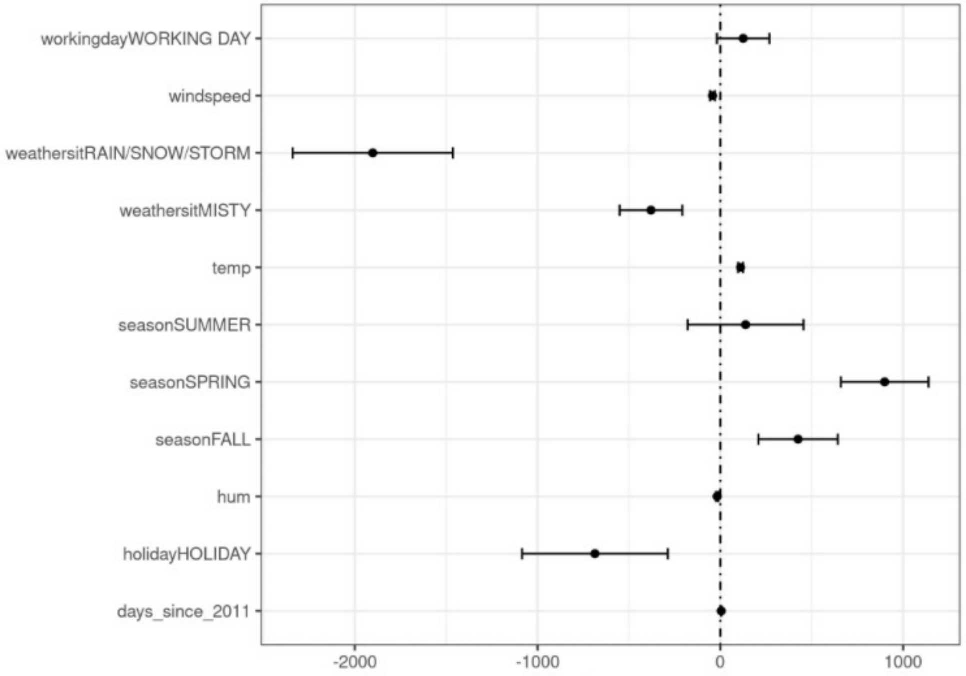
- larger weight → higher importance for the model
- Larger variance (less certain about the correct value) → lesser importance for the model

Interpretable Models
Linear Regression

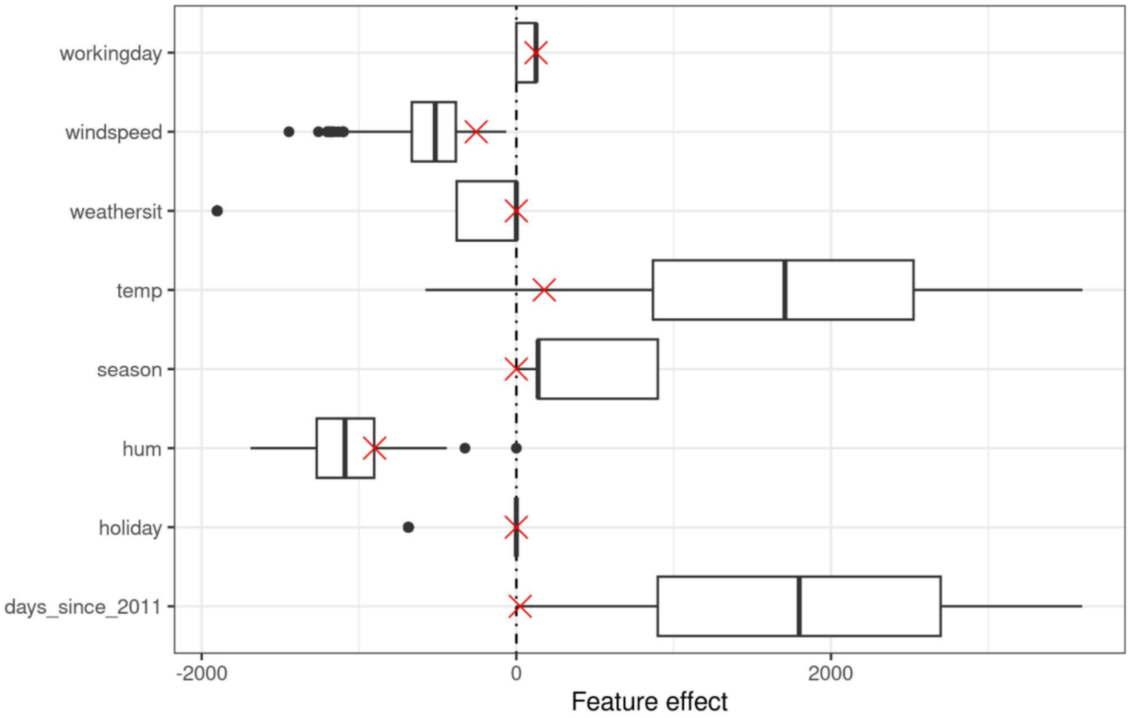


	Weight	SE	t-statistic
(Intercept)	2399.4	238.3	10.1
seasonSPRING	899.3	122.3	7.4
seasonSUMMER	138.2	161.7	0.9
seasonFALL	425.6	110.8	3.8
holidayHOLIDAY	-686.1	203.3	3.4
workingdayWORKING DAY	124.9	73.3	1.7
weathersitMISTY	-379.4	87.6	4.3
weathersitRAIN/SNOW/STORM	-1901.5	223.6	8.5
temp	110.7	7.0	15.7
hum	-17.4	3.2	5.5
windspeed	-42.5	6.9	6.2
days_since_2011	4.9	0.2	28.5

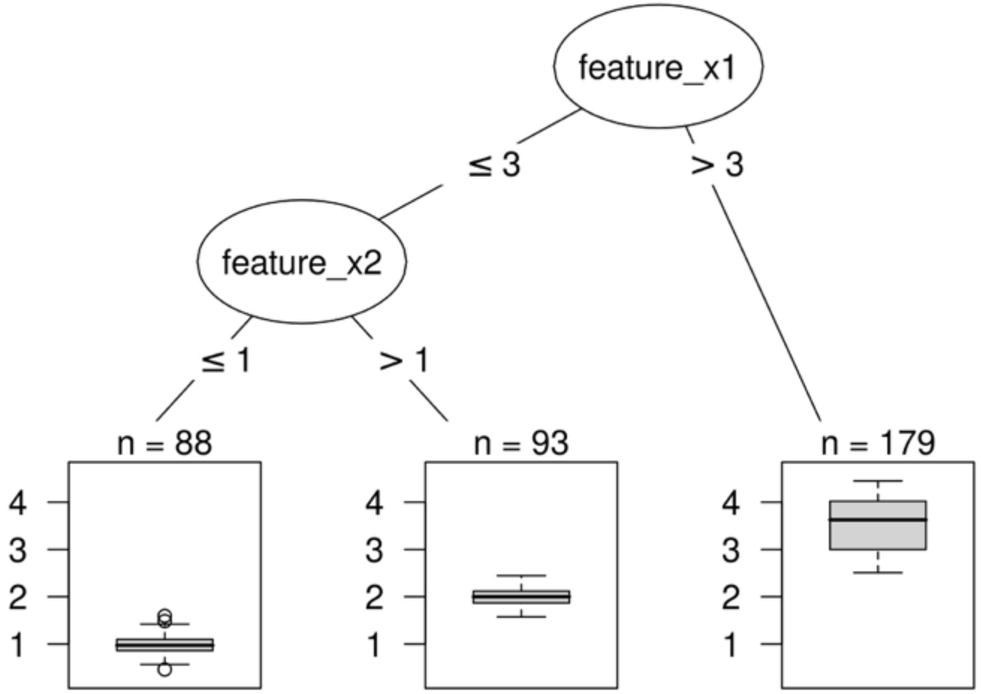
Interpretable Models
Linear Regression



$$\text{effect}_j^{(i)} = w_j x_j^{(i)}$$



Interpretable Models
Decision Trees

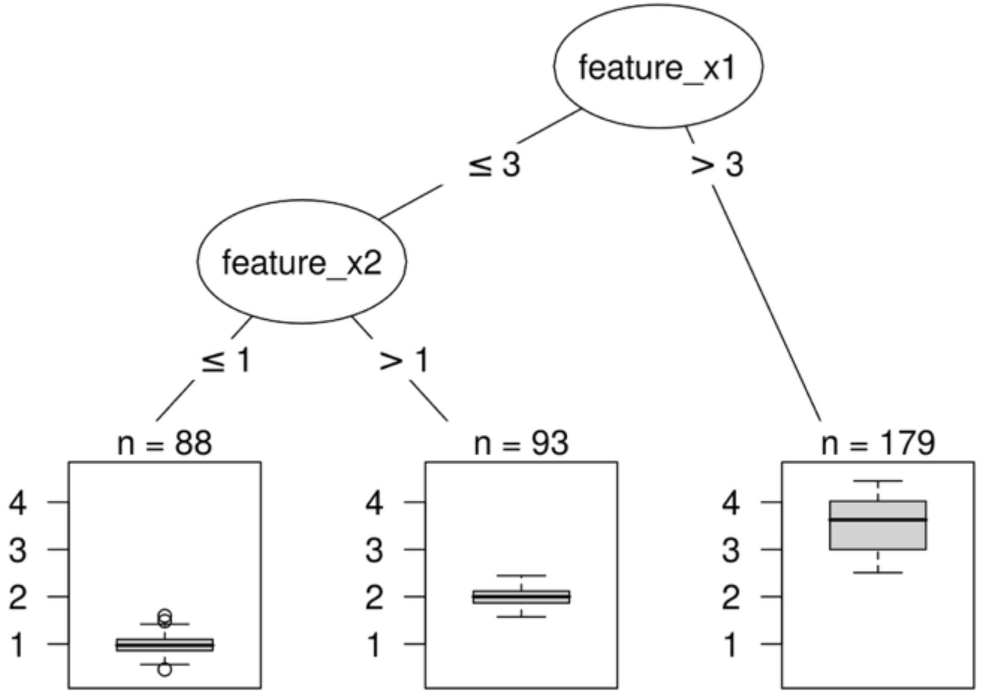


$$\hat{y} = \hat{f}(x) = \sum_{m=1}^{M} c_m I\{x \in R_m\}$$

Interpretable Models
Decision Trees

Classification and regression trees (CART)

- Regression → Splits on variance
- Classification → Splits on Gini Index

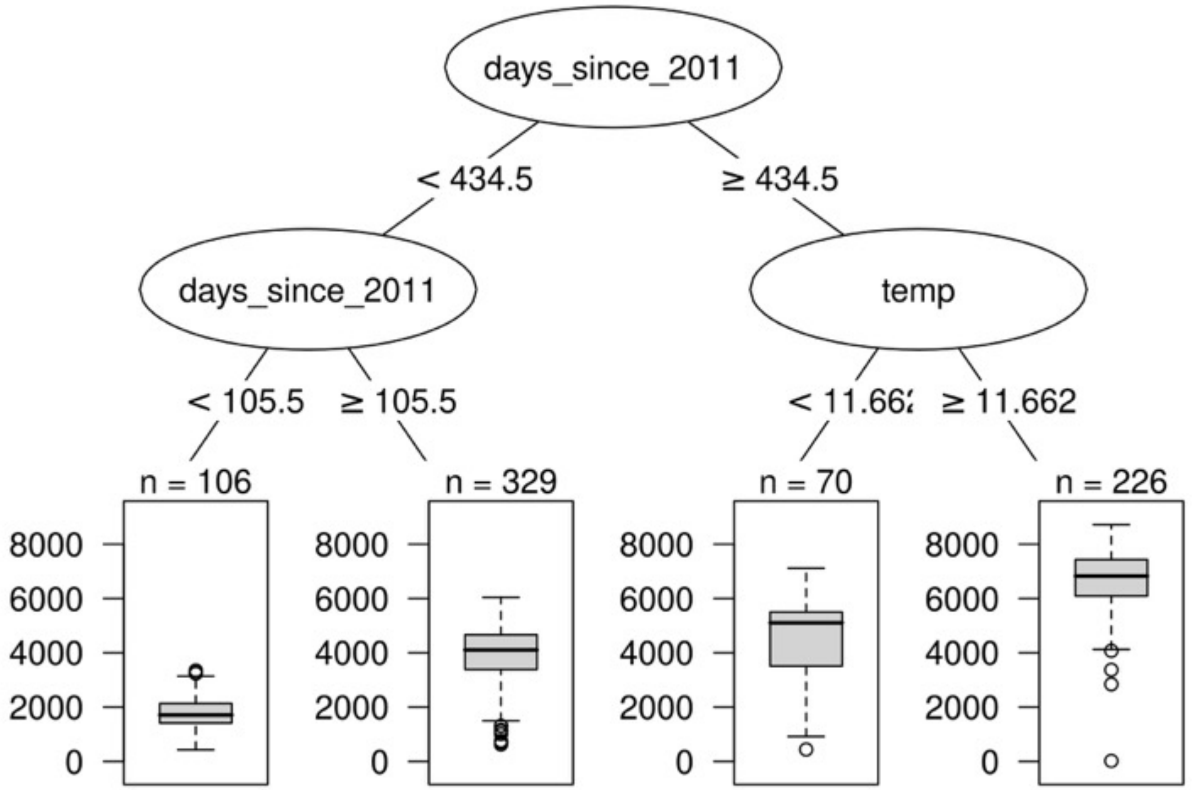


$$\hat{y} = \hat{f}(x) = \sum_{m=1}^{M} c_m I\{x \in R_m\}$$

Interpretable Models
Decision Trees

Interpretation

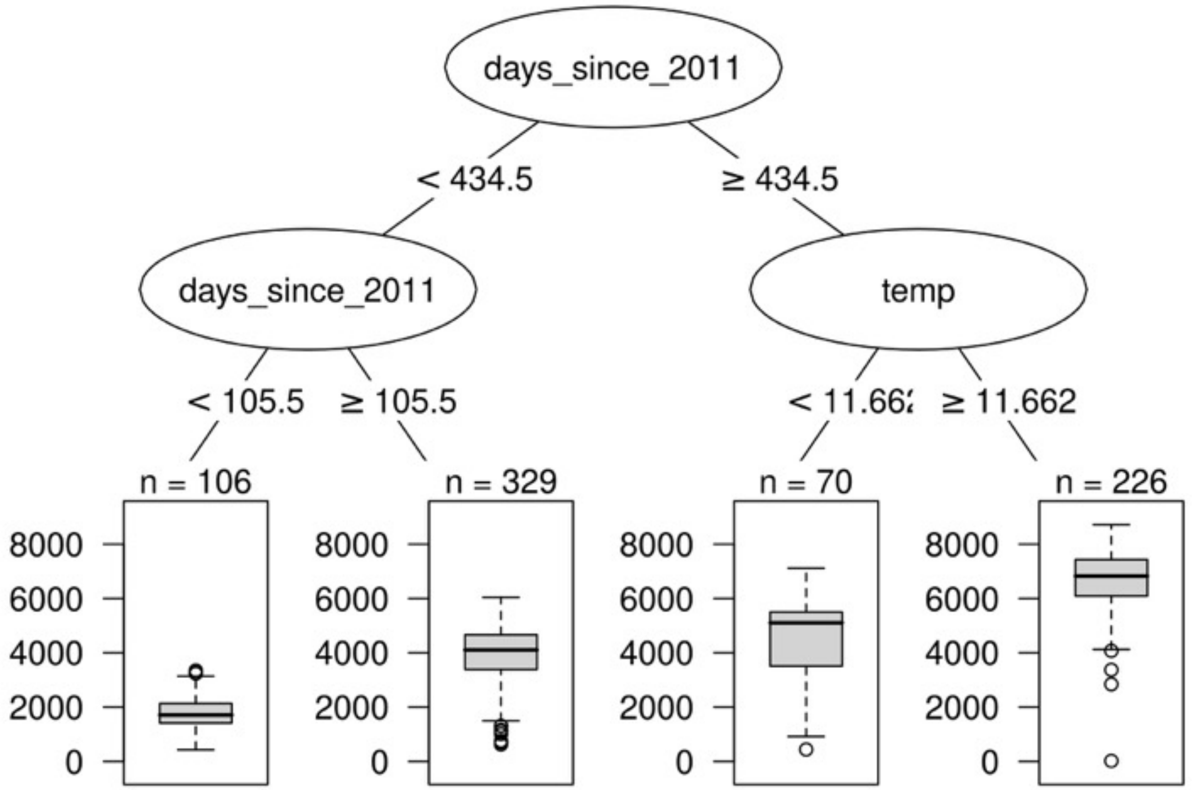
- If feature x is smaller/larger than threshold c
  AND …
  then the predicted outcome is the mean value of y of the instances in that node



Interpretable Models
Decision Trees

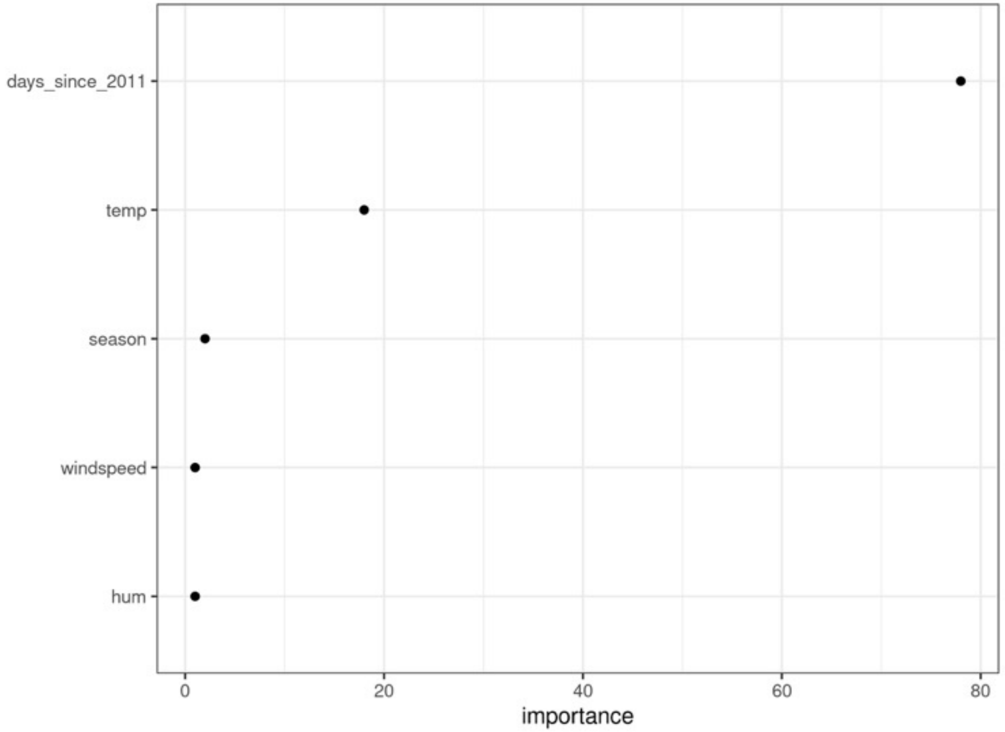
Feature Importance

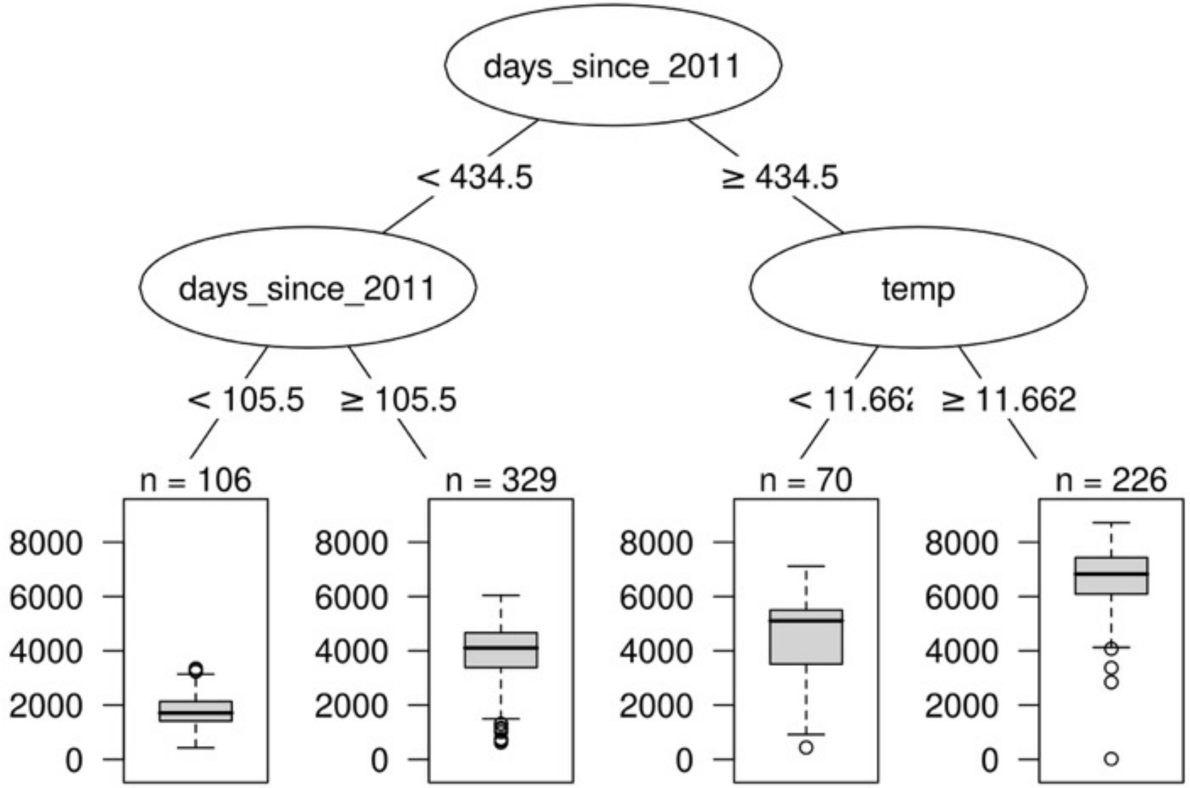
- Go through all splits for feature x and determine how much it reduced the variance/Gini compared to parent node



Interpretable Models
Decision Trees

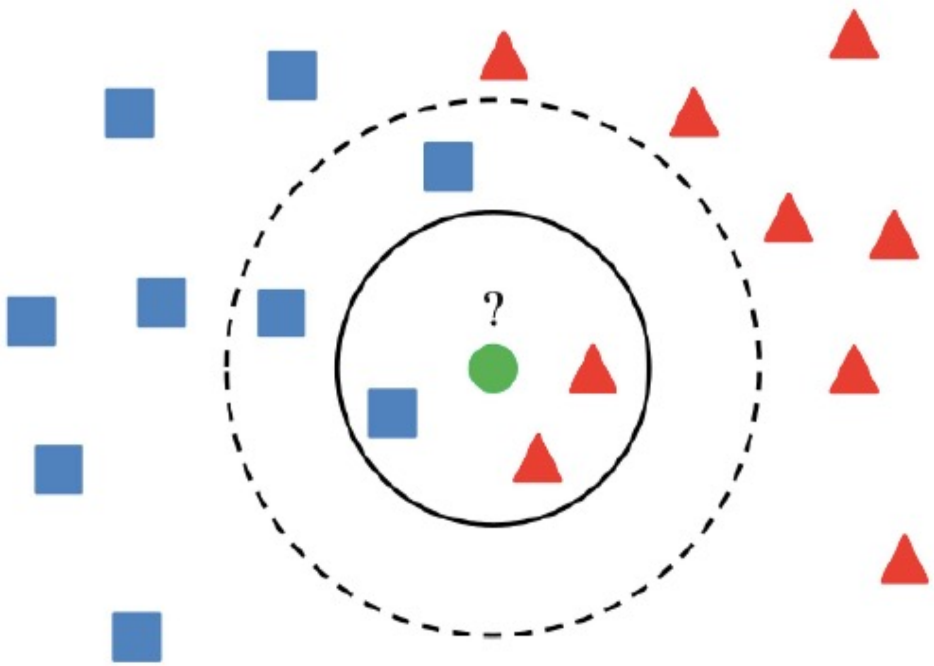
Feature Importance





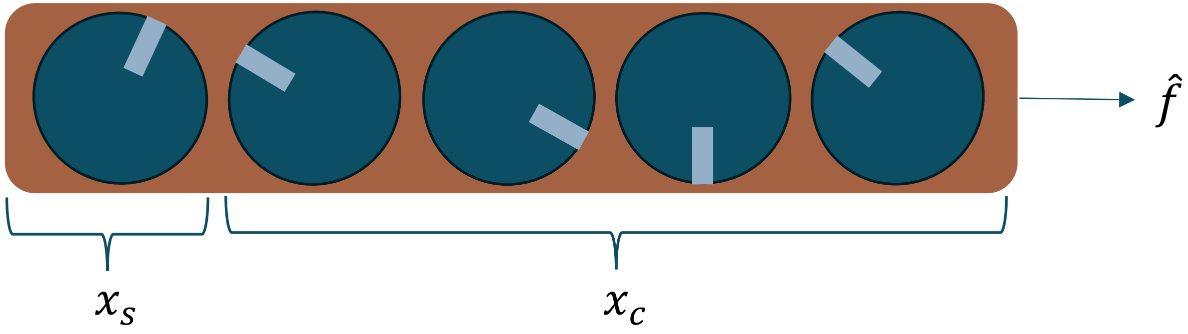
Interpretable Models
k-Nearest Neighbors

What do you think?



Post-hoc Methods

Global Model-Agnostic
Partial Dependence Plots (PDP)



For every feature value on x_s, we compute the average output for all known settings of x_c.

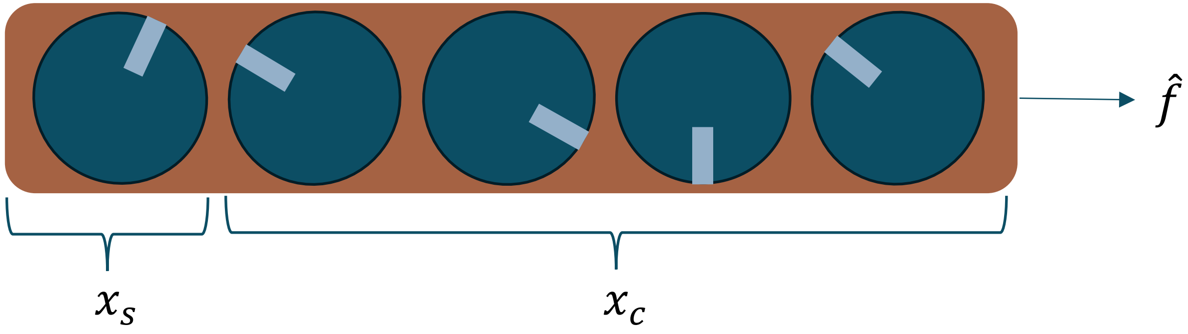
I.e, $\hat{f}_s(x_s) = \frac{1}{n}\sum_{i=1}^{n}\hat{f}\left(x_s, x_c^i\right)$

Shows how a particular feature relates to the output of a model

How do we turn these knobs to identify how x_s relates to output $\hat{f}$?

x_1	x_2	x_3	x_4	output
v_1	v_1	v_1	v_1	$\hat{f}$
v_1	v_2	v_2	v_2	$\hat{f}$
v_1	v_3	v_3	v_3	$\hat{f}$

Global Model-Agnostic
Partial Dependence Plots (PDP)



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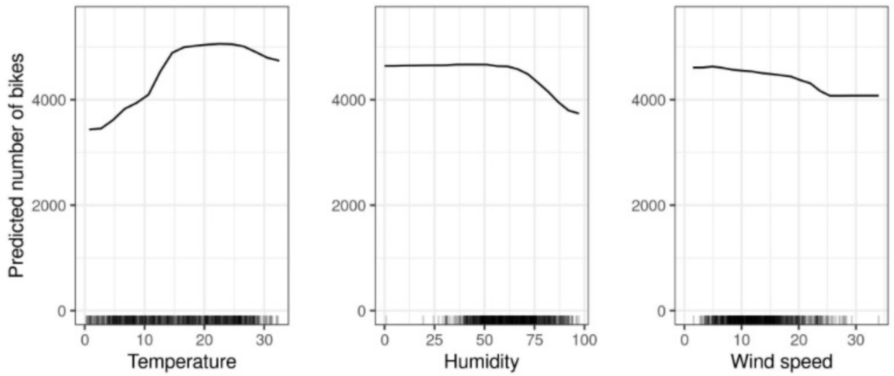
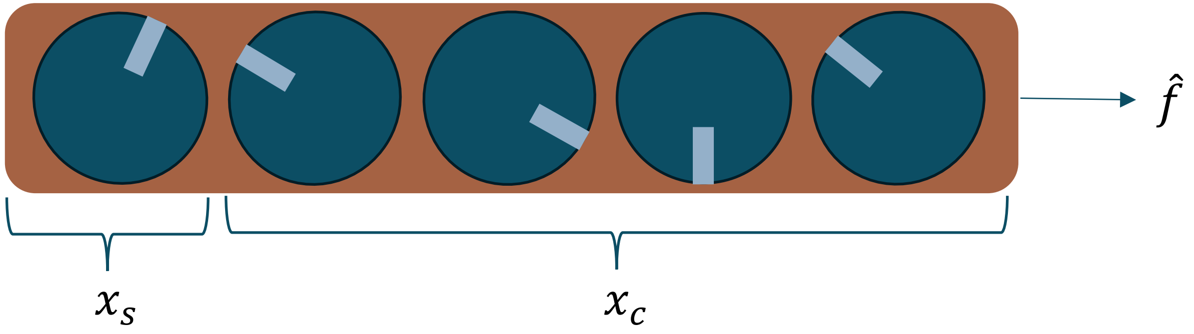


x_1	x_2	x_3	x_4	output
v_1	v_1	v_1	v_1	$\hat{f}$
v_1	v_2	v_2	v_2	$\hat{f}$
v_1	v_3	v_3	v_3	$\hat{f}$

Global Model-Agnostic
Partial Dependence Plots (PDP)



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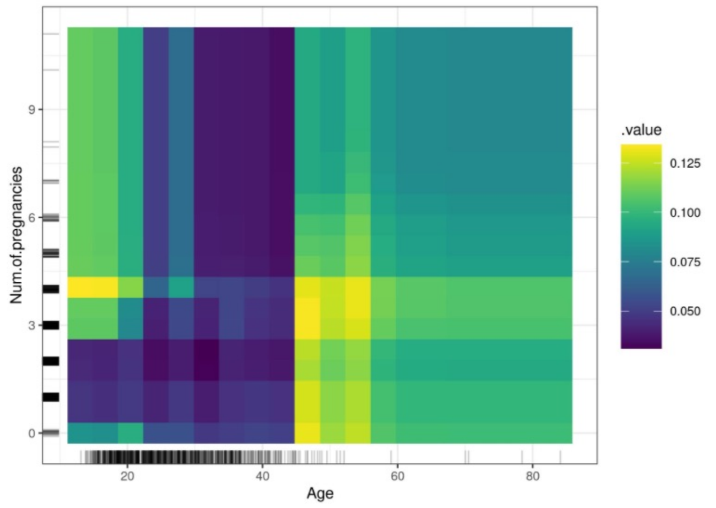
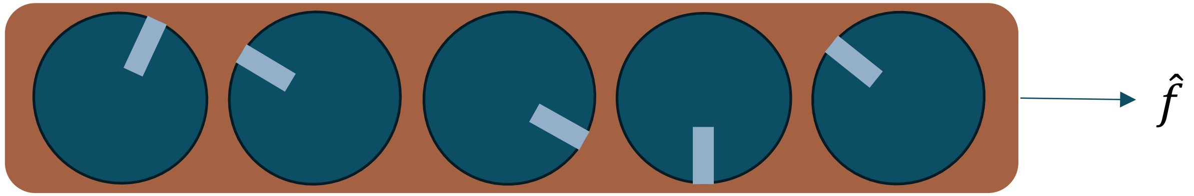


x_1	x_2	x_3	x_4	output
v_1	v_1	v_1	v_1	$\hat{f}$
v_1	v_2	v_2	v_2	$\hat{f}$
v_1	v_3	v_3	v_3	$\hat{f}$

Global Model-Agnostic
Permutation Feature Importance



Destroyer of relationships – And measure the effect by doing so

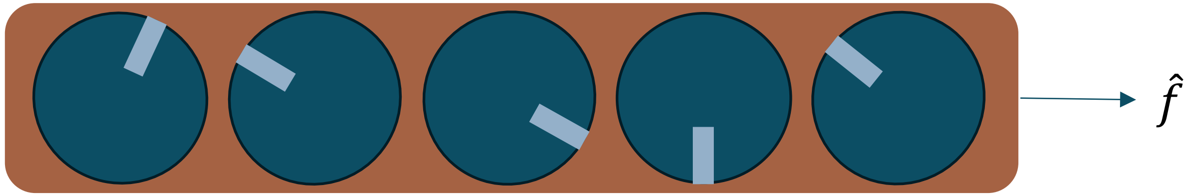
Scramble the values of each features individually

x_1	x_2	x_3	x_4	output
v_3	v_1	v_1	v_1	$\hat{f}$
v_1	v_2	v_2	v_2	$\hat{f}$
v_2	v_3	v_3	v_3	$\hat{f}$

Input: Trained model $\hat{f}$, feature matrix X, target vector y, error measure $L(y, \hat{f})$.

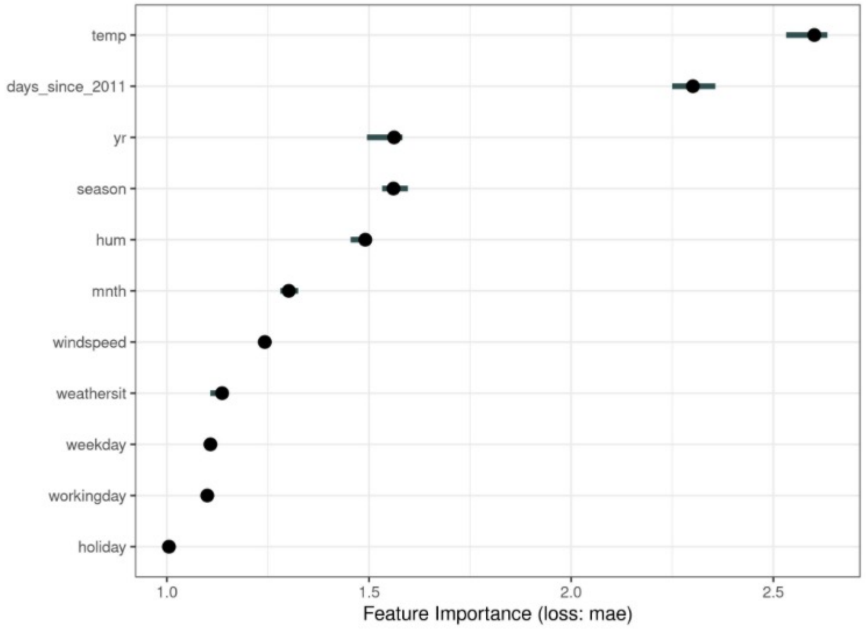
1. Estimate the original model error $e_{orig} = L(y, \hat{f}(X))$ (e.g. mean squared error)
2. For each feature $j \in \{1, \ldots, p\}$ do:
   - Generate feature matrix X_{perm} by permuting feature j in the data X. This breaks the association between feature j and true outcome y.
   - Estimate error $e_{perm} = L(Y, \hat{f}(X_{perm}))$ based on the predictions of the permuted data.
   - Calculate permutation feature importance as quotient $FI_j = e_{perm}/e_{orig}$ or difference $FI_j = e_{perm} - e_{orig}$
3. Sort features by descending FI.

Global Model-Agnostic
Permutation Feature Importance



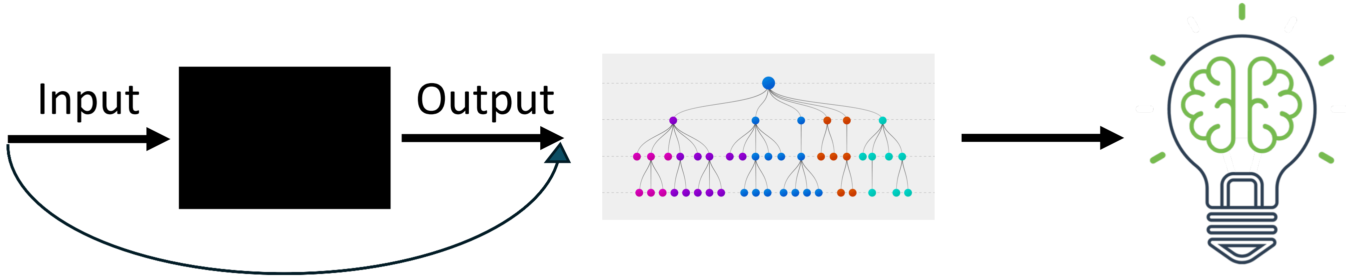
Destroyer of relationships – And measure the effect by doing so

Scramble the values of each features individually



Global Model-Agnostic
Global Surrogate

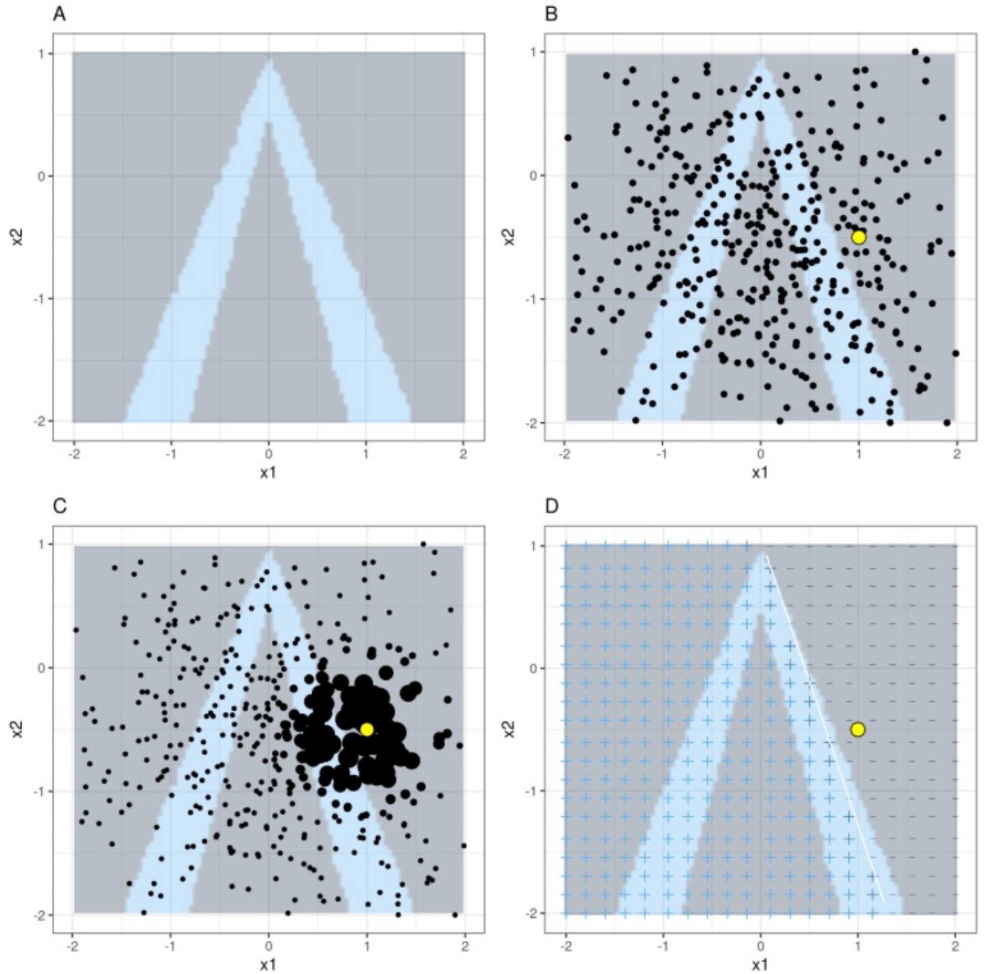
Use a less complex model to interpret the results from the black box



We get conclusions of the model, not the data.

Local Model-Agnostic
LIME

- Identify an area of interest in the model
- Generate a new dataset by pertubating the original data, and gather its corresponding predictions
- Weight the new instances according to their distance to the original point of interest
- Train surrogate model(s) on that data
- You now have an approximation of that part of the model



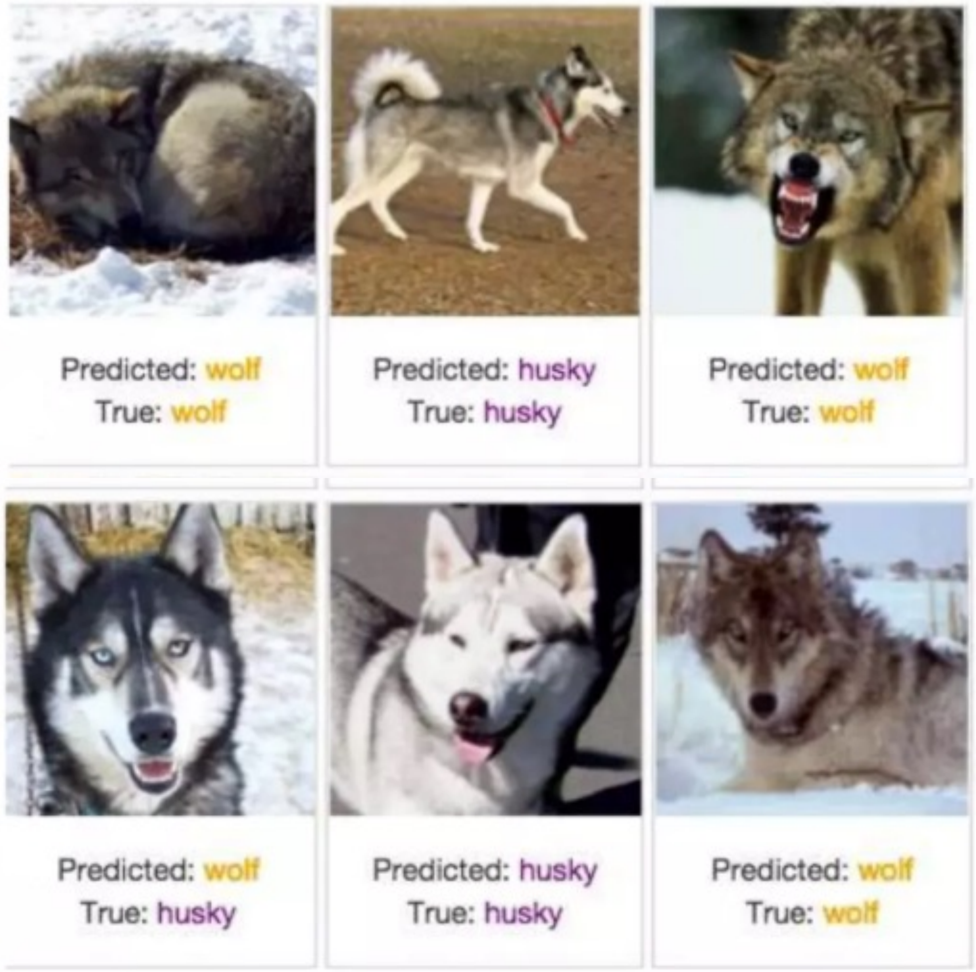
$$\text{explanation}(x) = \arg\min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

Local Model-Agnostic
LIME for images

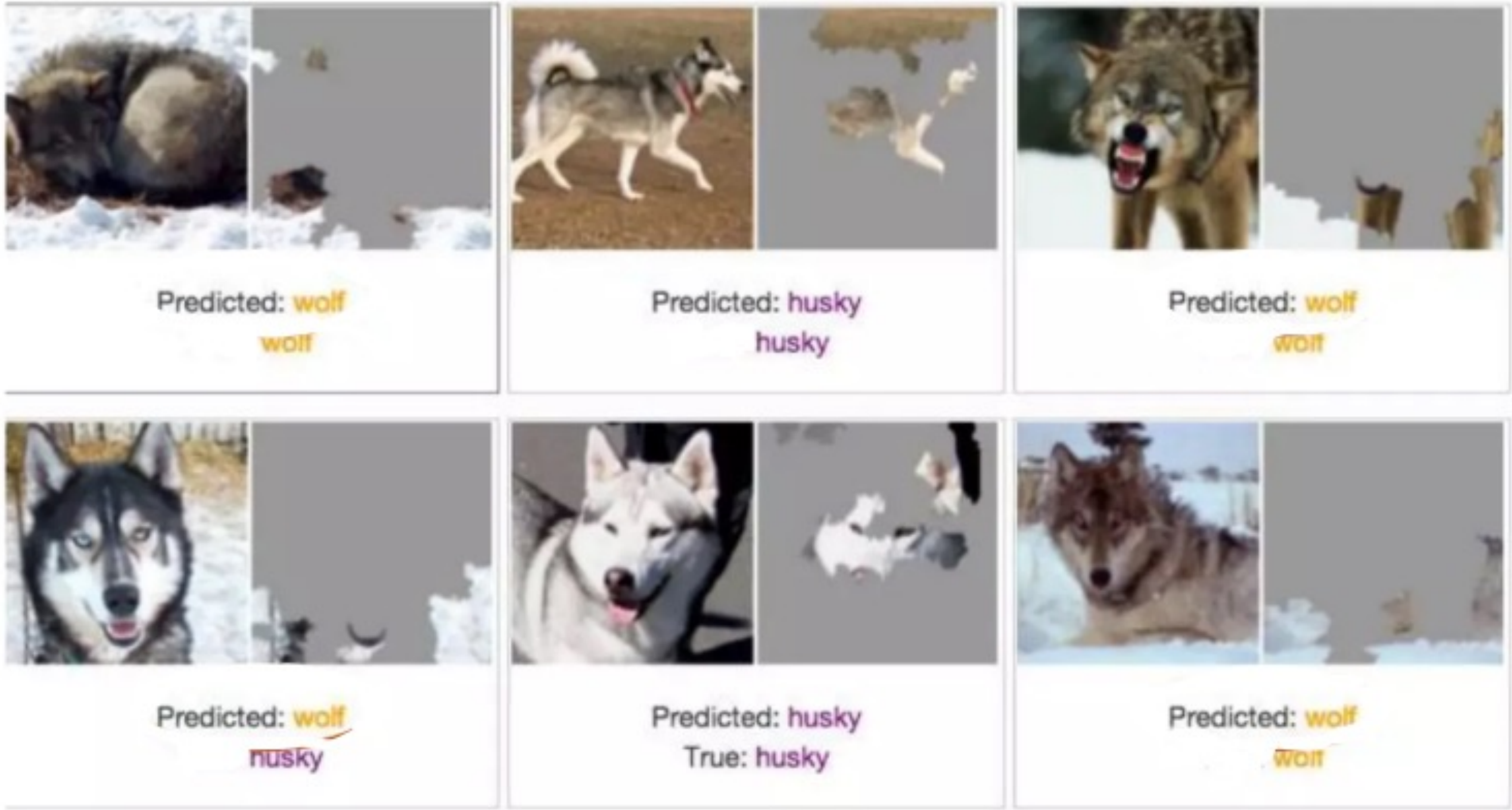
Can’t pertubate all features(pixels)

Segments into superpixels, interconnected pixels that are similar in color

Superpixels are turned off by changing color



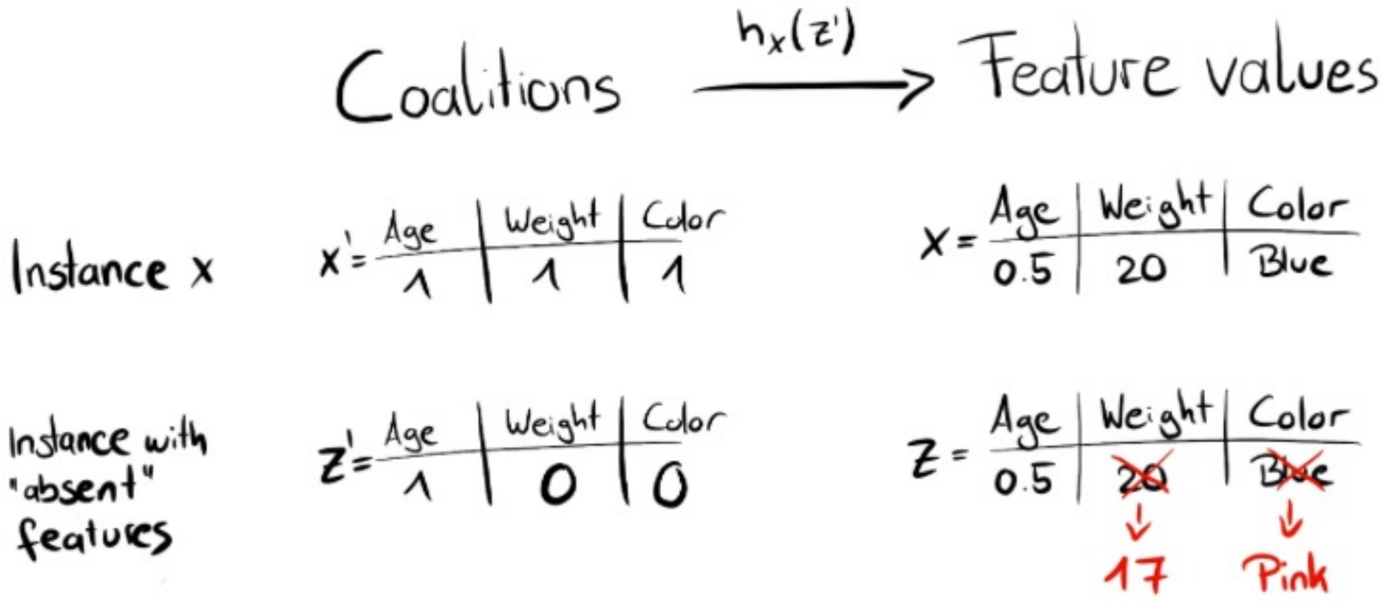
Local Model-Agnostic
LIME



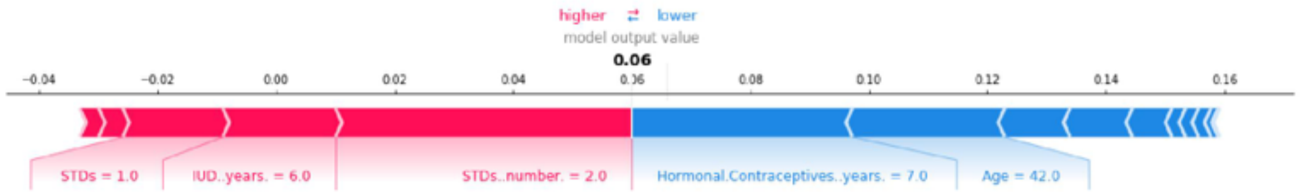
Local Model-Agnostic
SHAP

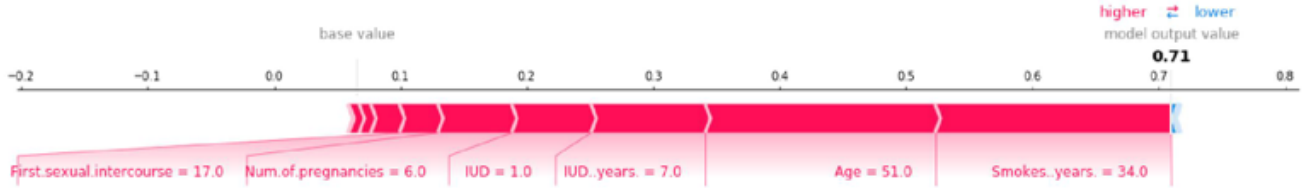
- Sample coalitions
- Gather predictions for each coalition
- Compute weight for each coalition
- Fit weighted linear model
- Calculate shapley values from the linear model

Local Model-Agnostic
SHAP

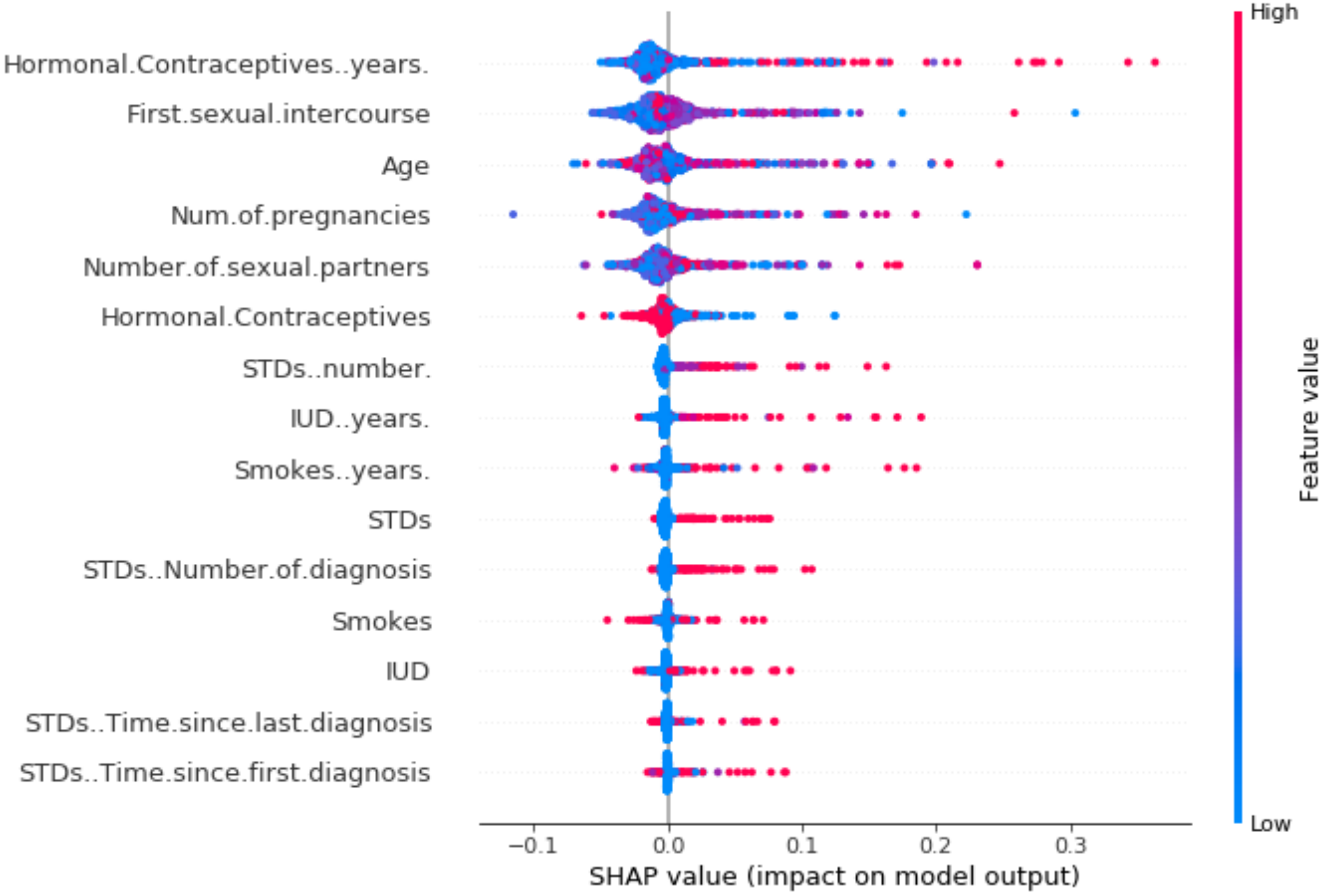


Local Model-Agnostic
SHAP





Local Model-Agnostic
SHAP



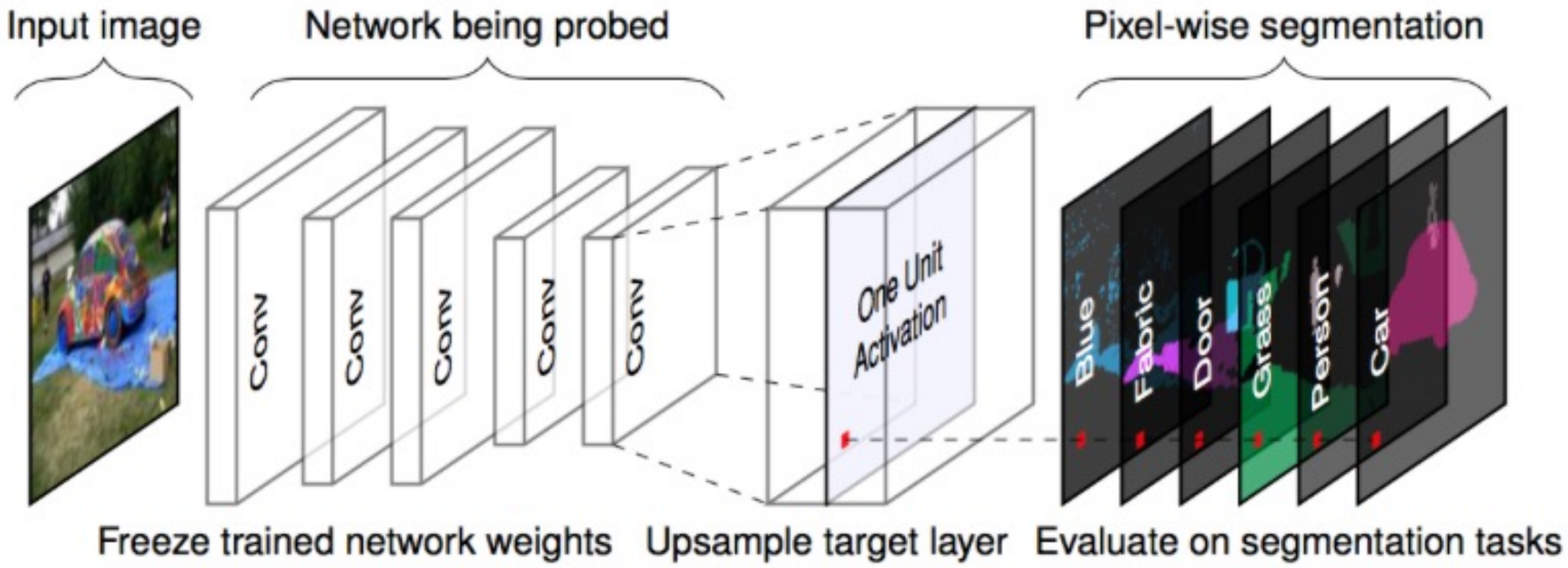
Local Model-Agnostic
Calibrated Explanations

Introduces uncertainty and other things to the local model-agnostic methods

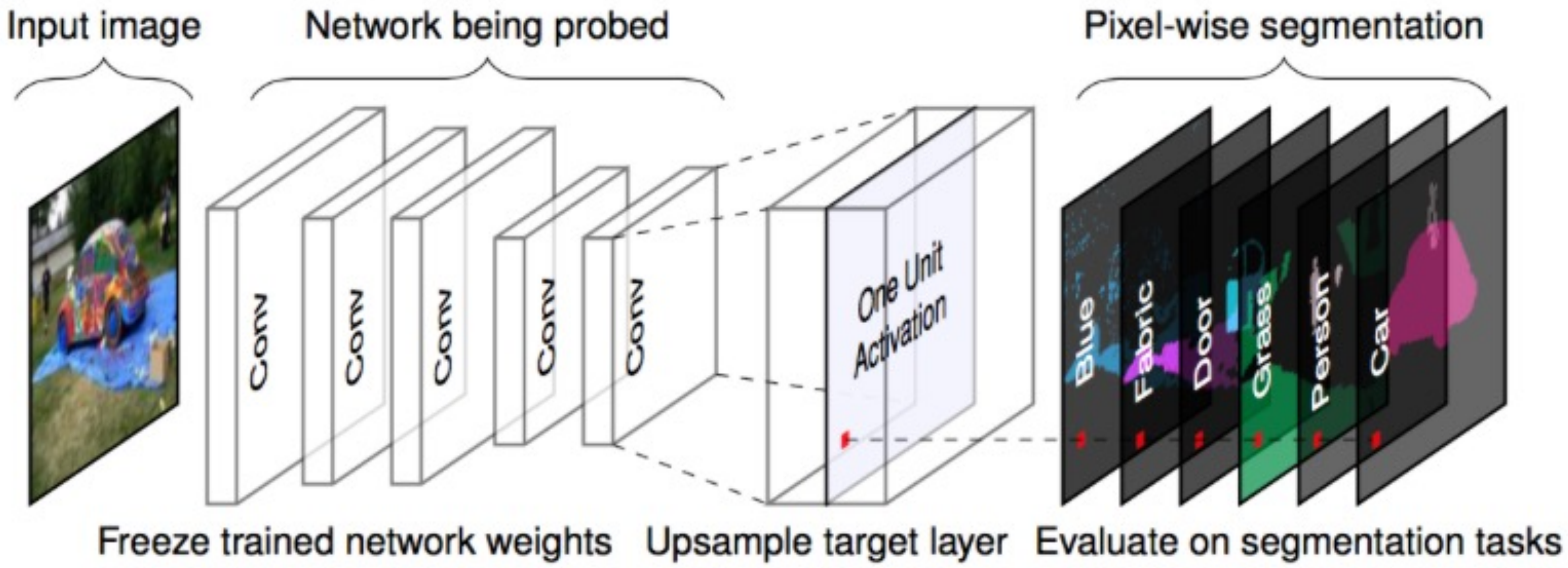
Thesis: Link_to_diva_record

Git: https://github.com/Moffran/calibrated_explanations

Neural Network Techniques
Learned Features – Network dissection

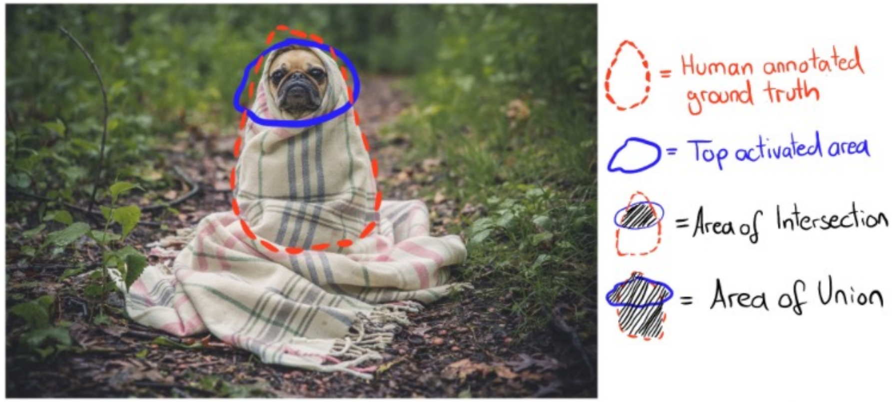


Neural Network Techniques
Learned Features – Network dissection



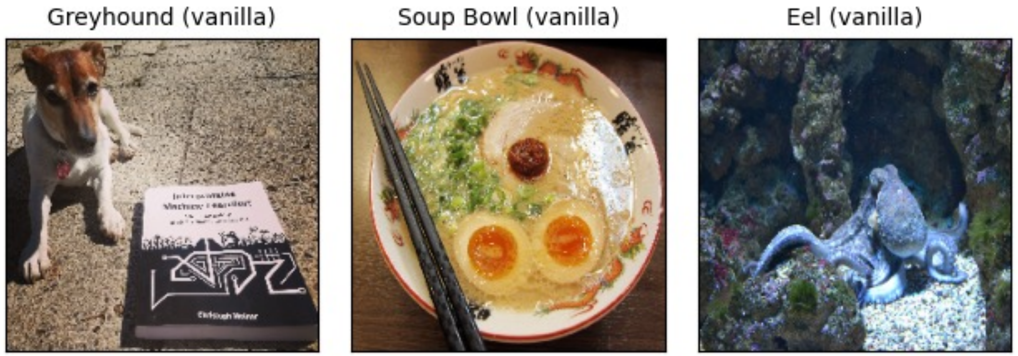
Neural Network Techniques
Learned Features – Network dissection

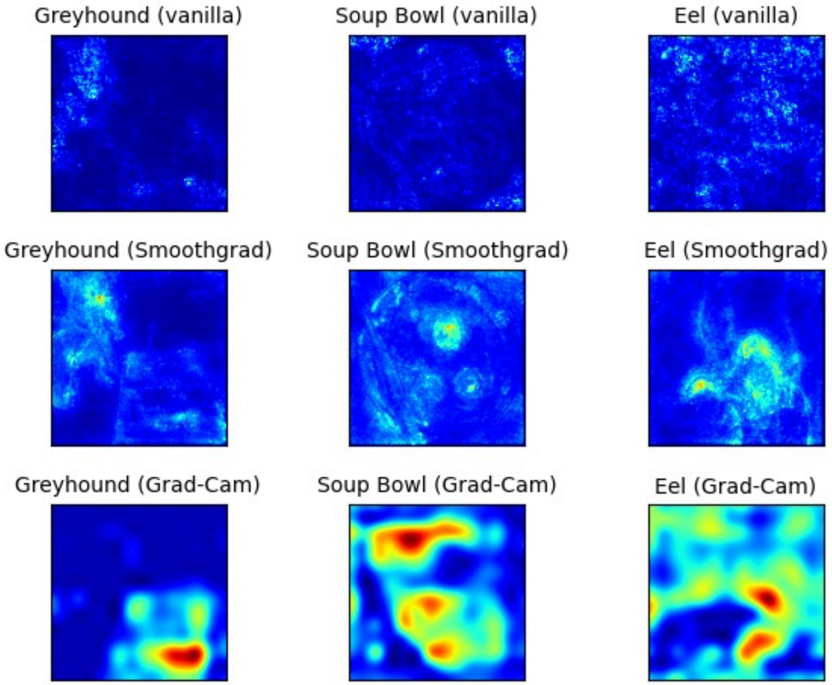
- For each convolutional channel k:
  - For each image x in the Broden dataset:
    - Forward propagate image x to the target layer containing channel k.
    - Extract the pixel activations of convolutional channel k: $A_k(x)$.
  - Calculate distribution of pixel activations α_k over all images.
  - Determine the 0.995-quantile level T_k of activations α_k. This means 0.5% of all activations of channel k for image x are greater than T_k.
  - For each image x in the Broden dataset:
    - Scale the (possibly) lower-resolution activation map $A_k(x)$ to the resolution of image x. We call the result $S_k(x)$.
    - Binarize the activation map: A pixel is either on or off, depending on whether or not it exceeds the activation threshold T_k. The new mask is $M_k(x) = S_k(x) \geq T_k(x)$.



Neural Network Techniques
Pixel Attribution

Identifies which pixels attributed the most to the final prediction





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